



ENTERPRISE ASSURANCE CONTROL PLANE

Measure trust before use.

Fuse evidence across domains and produce a provable decision before the enterprise restores, indexes, trains, deploys, or acts.

Can this digital asset be trusted before the enterprise acts on it, and can that decision be proven?

ALLOW

Trust sufficient

DENY

Block use

CONFIRM

Review evidence

ONE ASSURANCE MODEL

128D Behavioral Genome

8 SHARED ASSURANCE DIMENSIONS

Identity

Provenance

Behavior

Risk

Drift

Confidence

Evidence

Attestation

GX-GRA

Recovery

GX-GKA

Knowledge

GX-GMA

ML

GX-GAA

Agents

GX-GSA

Software

SHARED REPRESENTATION.
DOMAIN-SPECIFIC GATES.

GX-RA sits above the stack as the system of assurance.

Existing tools remain systems of record, detection, protection and delivery. GX-RA turns their fragmented signals into a governed trust decision at the moment of use.

Digital assets now participate in operations

A clean-looking asset can still carry stale, poisoned, compromised or unverifiable context. Availability and detection alone do not answer whether it should be used.



GX-RA APPLICATION LAYER

Operator console + industry views + domain gates

Sits on shared GenomeX engines. Consumes backup, EDR/XDR, AppSec, CI/CD, AI and observability evidence without replacing those platforms.

What GX-RA is not

Not a backup platform

Consumes backup and recovery state as evidence, then governs restore authorization.

Not EDR / XDR

Consumes Endpoint Detection & Response (EDR) and Extended Detection & Response (XDR) signals. It does not replace endpoint response.

Not an AppSec scanner

Fuses SAST / SCA findings, a Software Bill of Materials (SBOM), and release lineage into software trust.

Not a runtime LLM firewall

Governs upstream knowledge, datasets, agents and software before they enter the action path.

Not a GRC questionnaire

Produces operational evidence for

Instead: a decision control plane

OVERLAY, NOT REPLACEMENT

Every domain follows the same assurance discipline.

The asset changes. The before-use moment changes. The fusion and proof model remains consistent.



SHARED WIRE SCHEMA
platform_genome_128
One 128-dimensional genome representation, organized as 8 blocks × 16 slots. Domain workflows differ; evidence fusion reads the same blocks.

Engine	Obligation
GX-DAB Data Access Bridge	What data exists, where it lives, which subset may proceed.
GX-SEQ Sequencing Engine	Open the asset; catalog, scan, index, compare to baseline.
GX-SIG Signature lane	Known-bad signatures and indicators of compromise (IOCs).
GX-BE Behavior Engine	Baselines, drift, confidence and 128D behavioral genome.
GX-AL Assurance Ledger	Tamper-evident chain of custody and receipts.

FIVE ASSURANCE DOMAINS

Same model. Different before-use moment.

Enter the buyer through the domain where risk is most concrete, then expand across the same assurance substrate.

GX-GRA

GOVERNANCE RECOVERY ASSURANCE

Can this recovery chain be trusted?

GATE

Recovery Authorize / Time-Warp

PROOF / ARTIFACTS

BSR - Behavioral Snapshot Recovery

Merkle-verified recovery graph fusing Recovery Environment and Recovery Asset genomes.

PoS-B - Proof of State - Behavioral

Capture-time behavioral health attestation across the recovery chain.

GX-GKA

GOVERNANCE KNOWLEDGE ASSURANCE

Can this knowledge be trusted?

GATE

Pre-index governance

RECEIPT

KAR - Knowledge Assurance Receipt

Binds document hash, extraction, chunk manifest and policy decision.

GX-GMA

GOVERNANCE MACHINE LEARNING ASSURANCE

Can this ML asset be trusted?

GATE

Training / evaluation

RECEIPT

ML-BOM - Machine Learning Bill of Materials

Attests dataset hash, schema, licensing and policy state.

GX-GAA

GOVERNANCE AGENT ASSURANCE

Can this agent be trusted?

GATE

Action authorization

RECEIPT

AAR - Agent Assurance Receipt

Signed record of the evaluated tool action and gate decision.

GX-GSA

GOVERNANCE SOFTWARE ASSURANCE

Can this software be trusted?

GATE

Release authorization

EVIDENCE ARTIFACT

SBOM - Software Bill of Materials

Dependency evidence fused with provenance, build and release behavior.

Recovery is one domain, not the platform center.

GX-RA is the application layer for shared evidence, policy, operator workflow and proof across all five domains.

No silent pass/fail. Every gate is explicit.

Each outcome carries the evidence needed for operators, auditors, underwriters and service providers to understand what happened and why.

ALLOW

Trust is sufficient for this use. Proceed under policy and issue an assurance receipt.

DENY

Trust is insufficient. Block the use and preserve the evidence behind the decision.

CONFIRM

Evidence is incomplete, conflicting or policy-sensitive. Route to human or deeper review.

EVIDENCE FUSED AT DECISION TIME

Trust is not a single signal.

Discovery / scope

What exists, where it lives and what subset is eligible.

Materialized truth

What is actually inside the asset at policy depth.

Known-bad signals

Signatures, IOCs and threat intelligence matches.

Behavior + drift

Baseline deviation, confidence and contextual risk.

Attestation

Chain of custody, receipt and tamper-evident proof.

Domain policy

Use-specific rules that turn evidence into action.

Industry views: same evidence, different lens

CISO

CHIEF INFORMATION SECURITY OFFICER

Intervention rate, open CONFIRM decisions and domain posture.

GRC

GOVERNANCE, RISK & COMPLIANCE

Control coverage, attestations and export-ready audit packs.

Insurance

UNDERWRITING / RISK

Underwriting Assurance Receipt (UAR), risk posture and recovery-confidence evidence.

MSSP

MANAGED SECURITY SERVICE PROVIDER

Tenant rollup with green, amber and red posture at a glance.

From evidence to action, with proof built in.

GX-RA is the operational application surface on top of shared GenomeX engines, exposed through console, API, host telemetry and sequencing services.



GX-RA Operator console and industry views for domain posture, interventions and export.

gxra-api FastAPI control plane for authorize, policy, evidence and integration workflows.

gxra-agent Host telemetry source for behavior and state evidence.

GX-SEQ Universal sequencer for materialization and deep asset inspection.

BSR **Behavioral Snapshot Recovery:** Merkle-verified recovery graph and selective clean-ancestor restore evidence.

Trust becomes an evidence-backed decision.

GX-RA turns fragmented security, data, software and AI signals into one operational answer: **ALLOW**, **DENY** or **CONFIRM** - with the proof to defend it.

Receipt and acronym terminology aligned to GenomeX Assurance Platform White Paper v2.0, June 2026.